**Data Provided: Erlang C curves.**

**The University of Sheffield**

**DEPARTMENT OF COMPUTER SCIENCE**

**AUTUMN/SPRING SEMESTER 2002–2003** **2 HOURS**

**NETWORK PERFORMANCE ANALYSIS**

**Answer THREE questions.**

**All questions carry equal weight. Figures in square brackets indicate the percentage of available marks allocated to each part of a question.**

**1.** In an $M/G/1$ queue inter-arrival times are exponentially distributed and service times are 'generally' distributed.

a) Use the concept of residual service time to show that, assuming customers are served in the order that they arrive, the average waiting time is given by

$$\bar{W} = \frac{\bar{R}}{1-\rho},$$

where $\bar{R}$ is the average residual service time and $\rho$ is the queue utilization factor. Use diagrams, where necessary, to explain your reasoning and state your assumptions clearly.

[50%]

b) Use a graphical argument to derive a formula for the residual service time, then combine this equation with the formula for the average waiting time above to show that

$$\bar{W} = \frac{\lambda \langle X^2 \rangle}{2(1-\rho)},$$

where $\lambda$ is the arrival rate and $\langle X^2 \rangle$ is the expectation of the squared service time.

[40%]

c) The equation you have derived is easily recognized as the Pollaczek-Khinchin formula. Use it to derive the average waiting time in an $M/D/1$ queue.

[10%]

**2.** You are presented with the following portion of Tcl code designed for the network simulator NS2.

```
set node1 [$ns node]
set node2 [$ns node]
set node3 [$ns node]
set node4 [$ns node]

$node1 set multiPath_ 1

set link1 [$ns duplex-link $node1 $node2 1Kb 0ms DropTail]
set link2 [$ns duplex-link $node1 $node3 1Kb 0ms DropTail]
set link3 [$ns duplex-link $node2 $node4 1Kb 0ms DropTail]
set link4 [$ns duplex-link $node3 $node4 1Kb 0ms DropTail]

set src [new Agent/UDP]
$ns attach-agent $node1 $src
set sink [new Agent/Null]
$ns attach-agent $node4 $sink
```

a) Assuming that only the specified nodes exist, draw the network topology showing the associated routes.

[25%]

b) State the assumptions behind the Kleinrock Independence approximation. Discuss to what extent they are valid for this network topology.

[35%]

c) Packet arrivals occur as a Poisson distribution at `node1` with a rate of 2 per second. Packet lengths are distributed as an exponential distribution with a mean of 2 Kb. Assuming that packets are sent randomly along all possible routes with equal probability, use Kleinrock's independence assumption to estimate the steady state average number of packets that are at each link.

[25%]

d) What is the expected delay for each route through the network?

[15%]

**3. Hand in the Erlang C curves provided with your answer to this question.**

Northern Telecom has just completed the purchase of several local telecommunications companies, each with their own regional support teams. There are three regional support teams. The number of staff in each team and the call rate for each team's centre is given in the following table. You can assume that calls arrive as a Poisson process and that the service times are exponentially distributed.

| Region | A | B | C |
|---|---|---|---|
| $m$ | 10 | 13 | 10 |
| customers/hour | 120 | 160 | 100 |

a) Given that the average holding time, $\tau$, for support queries is 3 minutes, use the Erlang C curves at the back of the paper to give, for each region, the probability of queueing, $C(m,a)$.

[30%]

b) The formula for the average queue length in an $M/M/m$ system is

$$\bar{N}_Q = \frac{\rho}{1-\rho} C(m,a)$$

Use Little's formula to compute the formula for the average queueing time, and hence compute the expected queueing time for each call centre.

[30%]

c) Given that the average queueing time for customers who *don't* queue is zero. What is the average queueing time for customers who *do* queue?

[20%]

d) The company wishes to cut costs by merging the call centres to one location. Assuming that the total traffic level, the average holding time and the total staff numbers stay the same what will be the average queueing time for the new centre?

[20%]

**4.** A call centre employs thirty operators selling tickets for a major sporting event. The call centre has sixty lines in total. When all operators are active new calls are placed in a queue for the first available operator. The call centre is to be modelled as an $M/M/m/n$ queue, *i.e.* a queue in which there are $m$ servers and a maximum state of $n$.

a) Draw a state diagram representing the queueing system.

[30%]

b) Write down the detailed balance equations for the two cases where the queue state $k < m$ and when $m \leq k < n$.

[20%]

c) For each of the two cases outlined in the previous part write down the steady state distribution as a function of $P(0)$, the probability of the system being empty.

[30%]

d) Use the fact that the steady state distribution must be normalized to show that

$$P(0) = \left[ \sum_{k=0}^{m} \frac{a^k}{k!} + \sum_{k=m+1}^{n} \frac{a^k}{m! m^{k-m}} \right]^{-1}$$

where $a = \frac{\lambda}{\mu}$, $\lambda$ is the arrival rate and $\mu$ is the rate of the service time distribution.

[20%]

**END OF QUESTION PAPER**

**Registration Number** _____

